

Universal Comparison of Fabius Observation Masks

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Abstract

For independent bounded coordinates with summable caps, the experiment obtained by observing any coordinate mask under all total-sum reweightings is equivalent to observing the sum of that mask. Universal comparison of two masks is therefore characterized by probability convolution divisibility of their sum laws. The criterion permits overlapping masks, different cardinalities, countably many observations, and arbitrary independent additive noise. For common-tilt uniforms with reciprocal-integer geometric caps, it gives a complete order: every initial segment must contain at least as many source observations as target observations. For finite equal-cardinality masks with commensurate caps, the criterion becomes nonnegativity of the coefficients of an explicit polynomial quotient. Classical Gaussian polynomials give comparisons that cannot be realized by pairing individually divisible caps.

1 A convolution criterion for arbitrary masks

Let I be a finite initial segment of $\mathbb{N}$ or all of $\mathbb{N}$. Under a reference law Q , the coordinates X_i are independent and satisfy

$$0 \leq X_i \leq a_i, \quad a_i > 0, \quad A = \sum_{i \in I} a_i < \infty.$$

No density assumption is imposed in this section. Let Z be an arbitrary independent real random variable, and put

$$S = \sum_{i \in I} X_i, \quad T = S + Z.$$

The noise may be omitted by taking $Z = 0$. For Borel $w \geq 0$ with $0 < \mathbb{E}_Q w(T) < \infty$, define P_w by $dP_w/dQ = w(T)/\mathbb{E}_Q w(T)$.

For a mask $E \subseteq I$, define its vector, sum, and reference sum law by

$$V_E = (X_i)_{i \in E}, \quad U_E = \sum_{i \in E} X_i, \quad \mu_E = \text{Law}_Q(U_E).$$

Masks may be empty, finite, or countably infinite; the empty sum is zero and the empty vector has a one-point state space. Write $E^c = I \setminus E$.

We say that E *universally dominates* L if one Borel Markov kernel from V_E to V_L satisfies

$$\text{Law}_{P_w}(V_E)K = \text{Law}_{P_w}(V_L) \quad \text{for every admissible } w. \quad (1)$$

The kernel sees the source observation alone; it cannot see the total or the hidden noise. This is the Blackwell order for the experiments indexed by the weights w [1].

Theorem 1.1 (Arbitrary masks). *For any masks E, L , including overlapping masks of different cardinalities, universal dominance $E \rightarrow L$ is equivalent to*

$$\mu_E = \mu_L * \nu \quad (2)$$

for a probability measure ν . It is enough to require one kernel for all positive-probability events $T \leq t$, even only for rational t .

The factor ν is compactly supported and unique. Each vector experiment is equivalent to its sum experiment. The kernel between the sum experiments is unique μ_E -almost everywhere.

The final uniqueness statement concerns scalar sums. A vector kernel need not be unique. For any vector kernel, the uniquely determined scalar kernel is obtained by first averaging its input against the reference conditional law of V_E given U_E , then projecting its output to the target sum.

1.1 Reduction to the observed sum

The product spaces are compact metrizable, and U_E is continuous because its defining sum converges uniformly. Let

$$R_E(u, dv) = Q(V_E \in dv \mid U_E = u)$$

be a reference regular conditional probability. Unobserved coordinates and Z are independent of V_E , so conditioning on them does not change this conditional law. For bounded Borel f ,

$$\mathbb{E}[w(T)f(V_E)] = \mathbb{E}[w(T)R_E f(U_E)]. \quad (3)$$

The identity follows by conditional expectation for bounded weights and by truncation for every admissible weight. Hence the sum map and the reverse kernel R_E realize experiment equivalence. One reference version suffices, since every reweighted sum marginal is absolutely continuous with respect to μ_E .

Composing these maps shows that (1) holds exactly when a scalar kernel K satisfies

$$\mathbb{E}[w(T)Kf(U_E)] = \mathbb{E}[w(T)f(U_L)] \quad (4)$$

for all bounded Borel f and all weights. Thresholds suffice: multiplying each conditional identity by $Q(T \leq t)$ says that the finite signed measure

$$B \longmapsto \mathbb{E}[\mathbf{1}_{\{T \in B\}}(Kf(U_E) - f(U_L))]$$

vanishes on every lower half-line. Rational thresholds determine all half-lines by continuity from above. Uniqueness of finite measures, followed by integration and truncation of w , gives (4). Thresholds of probability zero give the zero identity automatically.

1.2 Proof of the convolution criterion

For each mask write $F_E(z) = \mathbb{E}e^{zU_E}$, an entire function. For bounded complex f , put

$$A_f(z) = \mathbb{E}[e^{zU_E}Kf(U_E)], \quad B_f(z) = \mathbb{E}[e^{zU_L}f(U_L)].$$

By independence of each mask and its own complement, (4) gives

$$\phi_Z(t)F_{E^c}(it)A_f(it) = \phi_Z(t)F_{L^c}(it)B_f(it), \quad t \in \mathbb{R},$$

where $\phi_Z(t) = \mathbb{E}e^{itZ}$. Since $\phi_Z(0) = 1$, it is nonzero near zero. Cancel it there, and use analyticity of the remaining compact transforms to obtain

$$F_{E^c}(z)A_f(z) = F_{L^c}(z)B_f(z) \quad (z \in \mathbb{C}). \quad (5)$$

No exponential moment of Z is required. Moreover,

$$F_E F_{E^c} = F_I = F_L F_{L^c}. \quad (6)$$

All transforms equal one at zero. Dividing (5) and (6) near zero yields

$$A_f(z) = \frac{F_E(z)}{F_L(z)} B_f(z). \quad (7)$$

For necessity, take $X \sim \mu_E$ and sample Y conditionally from $K(X, dy)$. The reference weight gives $Y \sim \mu_L$. Put $D = X - Y$, a bounded random variable. For bounded g , substitute $f(y) = e^{-ity}g(y)$ in (7), for small real t :

$$\mathbb{E}[e^{itD}g(Y)] = \frac{F_E(it)}{F_L(it)} \mathbb{E}g(Y).$$

Taking $g = 1$ identifies the quotient as $\mathbb{E}e^{itD}$. The entire function

$$z \mapsto \mathbb{E}[e^{zD}g(Y)] - \mathbb{E}[e^{zD}]\mathbb{E}g(Y)$$

therefore vanishes on an imaginary interval and hence everywhere. Taking g to be indicators and using Fourier uniqueness proves independence of D, Y . This gives (2) with $\nu = \text{Law}(D)$.

Conversely, assume (2). Choose independent $Y \sim \mu_L$ and $D \sim \nu$, and set $X = Y + D$. The compact laws of X, Y force D to be bounded. Let $K = \text{Law}(Y | X)$. Equations (2) and (6), with cancellation of F_L near zero and analytic continuation, imply

$$\mu_{L^c} = \nu * \mu_{E^c}. \quad (8)$$

Take $B \sim \mu_{E^c}$ independent of (Y, D) and of the prescribed noise Z . Then

$$\mathbb{E}[w(X + B + Z)Kf(X)] = \mathbb{E}[w(Y + D + B + Z)f(Y)]. \quad (9)$$

The left side is the source sum experiment, and (8) identifies the right side as the target sum experiment. Composing with the conditional vector kernels proves the asserted mask comparison.

For scalar-kernel uniqueness, subtract (5) for two kernels. Since F_{E^c} is nonzero near zero, the compact transform of $(K_1f - K_2f)\mu_E$ vanishes identically. Fourier uniqueness gives equality for each f almost everywhere. A countable family of rational lower-half-line indicators supplies one common null set on which to discard exceptions. Factor uniqueness follows by cancelling F_L near zero; every possible factor is compact as above. This completes the proof of Theorem 1.1.

Corollary 1.2 (Common observations cancel). *Put $C = E \cap L$, $E_0 = E \setminus C$, and $L_0 = L \setminus C$. Universal $E \rightarrow L$ dominance is equivalent to universal $E_0 \rightarrow L_0$ dominance. When it holds, there is a realizing kernel which retains V_C exactly, including for countably infinite C .*

Proof. The identities $\mu_E = \mu_C * \mu_{E_0}$ and $\mu_L = \mu_C * \mu_{L_0}$ allow cancellation of F_C near zero. Thus

$$\mu_E = \mu_L * \nu \iff \mu_{E_0} = \mu_{L_0} * \nu.$$

Apply the preceding construction to E_0, L_0 in the coordinate family with C removed. It gives equality jointly with that family's total. The vector V_C is independent before reweighting; retain it and add its sum to the total. The joint identity, multiplied by $w(T)$, proves the claim. $\square$

2 The complete order for geometric caps

Assume now that the coordinate densities have a common real tilt:

$$q_{\beta,a}(x) = \frac{e^{-\beta x}}{Z_\beta(a)} \mathbf{1}_{(0,a)}(x), \quad Z_\beta(a) = \int_0^a e^{-\beta u} du. \quad (10)$$

The case $\beta = 0$ is the uniform law. Suppose $a_i = cM^{-i}$, where $c > 0$ and $M \geq 2$ is an integer. For a mask define

$$N_E(j) = |E \cap \{1, \dots, j\}|.$$

Theorem 2.1 (Prefix-count classification). *For arbitrary finite or countable masks, universal dominance $E \rightarrow L$ holds if and only if*

$$N_E(j) \geq N_L(j) \quad \text{for every } j \in I. \quad (11)$$

Equivalently, every target index ℓ_r in the increasing enumeration of L has a source index e_r in the increasing enumeration of E with $e_r \leq \ell_r$. Any additional source observations can be discarded. A realizing map is

$$(x_i)_{i \in E} \mapsto (x_{e_r} \bmod a_{\ell_r})_r. \quad (12)$$

The necessity permits every randomized kernel that combines source coordinates; it is not restricted to maps acting separately on coordinates.

Proof. The coordinate transform is

$$F_i(z) = \frac{e^{(z-\beta)a_i} - 1}{(z - \beta)Z_\beta(a_i)},$$

with its removable value at $z = \beta$. At

$$z_j = \beta + \frac{2\pi i}{a_j},$$

it has a simple zero exactly when $i \leq j$: the ratio $a_i/a_j = M^{j-i}$ is then an integer, whereas for $i > j$ it lies strictly between zero and one. Hence F_E has zero multiplicity $N_E(j)$ at z_j .

For an infinite mask the last assertion requires checking the tail product. Uniformly on any compact disk,

$$|F_i(z) - 1| \leq |z|a_i e^{|z|a_i}. \quad (13)$$

The products converge locally uniformly because $\sum_i a_i < \infty$, and agree with the transforms of the uniformly convergent coordinate sums. On a neighborhood of a fixed z_j , all sufficiently late factors lie within distance $1/2$ of one. Their analytic logarithms converge uniformly by (13), so the tail product is nonzero. Among the finitely many remaining factors, exactly the stated simple zeros contribute. Under dominance, $F_E = F_L F_\nu$ by Theorem 1.1. Zero multiplicities now force (11).

For sufficiency, pair the increasing indices $e_r \leq \ell_r$. The integer cap ratio allows the decomposition

$$X_{e_r} = a_{\ell_r} J_r + R_r, \quad R_r = X_{e_r} \bmod a_{\ell_r}.$$

Factoring the density in (10) shows that J_r and R_r are independent and $R_r \stackrel{d}{=} X_{\ell_r}$. Independence of source coordinates makes the entire residue vector R independent of all the digits and all unused source coordinates. For countably many coordinates this follows on determining cylinder sets. Set

$$D = \sum_r a_{\ell_r} J_r + \sum_{i \in E \setminus \{e_r : r\}} X_i.$$

These nonnegative sums converge and are bounded by $\sum_{i \in E} a_i$. Thus

$$U_E = \sum_r R_r + D, \quad D \text{ is independent of } R, \quad \text{Law}(R) = \text{Law}(V_L).$$

This proves $\mu_E = \mu_L * \text{Law}(D)$. It also verifies the deterministic map itself. By (8), the independent variable $D + U_{E^c}$ has law μ_{L^c} , so the joint laws of (T, R) and (T, V_L) coincide. Reweighting by $w(T)$ proves (12), even with overlap or infinitely many outputs. No infinite sequence of descending swaps is needed.

Finally, (11) forces at least r source indices by the r th target index. Conversely, each target index at most j has a matched source index at most j . This proves the equivalence with the sorted-index condition, including empty masks. $\square$

For the Fabius convention $a_i = t_n q^i$, $q = 1/M$, the map in unit coordinates is $u_{e_r} \mapsto \{M^{\ell_r - e_r} u_{e_r}\}$. The earlier sufficient comparison [6] is therefore optimal for this full family of experiments, even when arbitrary randomized pooling of observations is allowed.

3 An exact polynomial test for commensurate caps

The following different specialization gives comparisons beyond separately divisible coordinate pairs. It assumes finite masks of the same cardinality.

Let source caps be $ha_1, \dots, ha_k$ and target caps $hb_1, \dots, hb_k$, with $h > 0$, $k \geq 1$, and all a_i, b_i positive integers. Keep the common tilt $\beta \in \mathbb{R}$ from (10). Define

$$[n]_u = 1 + u + \dots + u^{n-1}, \quad P(u) = \frac{\prod_{i=1}^k [a_i]_u}{\prod_{i=1}^k [b_i]_u}, \quad r = e^{-\beta h}. \quad (14)$$

Theorem 3.1 (Polynomial criterion). *Universal source-to-target dominance holds if and only if the rational function P in (14) is a polynomial with nonnegative coefficients. When it is a polynomial its coefficients are integers and its constant coefficient is one. Writing $P(u) = \sum_j p_j u^j$, the unique convolution factor is*

$$\nu_\beta = \frac{1}{P(r)} \sum_j p_j r^j \delta_{hj}. \quad (15)$$

Thus the criterion does not depend on the common tilt. Masks may overlap.

Proof. The block decomposition of an interval gives the transform identity

$$F_{\beta, hn}(z) = F_{\beta, h}(z) \frac{[n]_{e^{h(z-\beta)}}}{[n]_{e^{-\beta h}}}. \quad (16)$$

Both masks have k coordinates, so the factors $F_{\beta, h}(z)^k$ cancel. Near zero,

$$\frac{F_E(z)}{F_L(z)} = \frac{P(e^{h(z-\beta)})}{P(r)}. \quad (17)$$

Suppose dominance holds. Write $N(u) = \prod_i [a_i]_u$ and $D(u) = \prod_i [b_i]_u$. Every zero ω of D is a root of unity different from one. Choose z_0 with $e^{h(z_0-\beta)} = \omega$. The factor $F_{\beta, h}(z_0)$ is nonzero, since its numerator is $\omega - 1$ and $z_0 - \beta \neq 0$. The exponential map has nonzero derivative $h\omega$. Entire-transform divisibility $F_E = F_L F_\nu$ therefore forces N to vanish at ω with at least the multiplicity of D . It follows that D divides N as polynomials. Monic integer polynomial division gives $P \in \mathbb{Z}[u]$, and $P(0) = 1$.

Equation (17) now identifies the transform of ν with that of the finite signed measure in (15). Its normalization is positive because $P(r) = N(r)/D(r) > 0$. Fourier uniqueness identifies the measures themselves. The sites hj are distinct, so positivity of ν forces $p_j \geq 0$. Conversely, if all coefficients are nonnegative, (15) is a probability measure and (17) proves $\mu_E = \mu_L * \nu_\beta$. Apply Theorem 1.1. $\square$

An explicit kernel independent of the tilt. Let g_L be the density of the sum of independent ordinary uniforms on the target caps. Given source sum s , choose j with probabilities

$$\Pr(j \mid s) = \frac{p_j g_L(s - hj)}{\sum_v p_v g_L(s - hv)}. \quad (18)$$

Set the target sum to $s - hj$, then draw the target vector from its ordinary-uniform conditional distribution given that sum. Set arbitrary values where the denominator vanishes; that set is source-null. To check the formula under tilt β , multiply the factor mass $p_j e^{-\beta h j} / P(r)$ by the tilted target-sum density at $s - hj$. The common exponential factor is $e^{-\beta s}$, which cancels from the posterior. The target vector's conditional law given its sum is also unchanged by the common tilt. Thus this same kernel works for every real β and every admissible total-sum weight.

Equal cardinality matters in Theorem 3.1: without it an additional power of the base-cap transform remains in (17). The general convolution criterion still applies, but the stated coefficient test is not asserted for that wider situation.

4 Gaussian polynomials and a positivity obstruction

For integers $n \geq k \geq 1$, take source caps

$$h(n - k + 1), \dots, hn$$

and target caps $h, 2h, \dots, kh$. Then P is the classical Gaussian polynomial

$$G_{n,k}(u) = \prod_{j=1}^k \frac{[n - k + j]_u}{[j]_u}. \quad (19)$$

Its coefficients are nonnegative integers. For completeness this follows inductively from the elementary recurrence

$$G_{n,k} = G_{n-1,k} + u^{n-k} G_{n-1,k-1}, \quad G_{n,0} = G_{n,n} = 1.$$

The product identity and the interpretation by partitions in a rectangle are classical [4, 5]; no novelty is claimed for them. Theorem 3.1 converts their positivity into universal mask comparisons. At $\beta = 0$, the factor law normalizes by $G_{n,k}(1) = \binom{n}{k}$.

For example, source caps (5, 6, 7) and target caps (1, 2, 3) give

$$\begin{aligned} G_{7,3}(u) = & 1 + u + 2u^2 + 3u^3 + 4u^4 + 4u^5 + 5u^6 \\ & + 4u^7 + 4u^8 + 3u^9 + 2u^{10} + u^{11} + u^{12}. \end{aligned}$$

The coefficients sum to 35. Universal garbling exists, although there is no bijection pairing each target cap with an integer-divisible source cap: both 2 and 3 would need the source cap 6. Formula (18) implements the comparison through the pooled source sum.

In the opposite direction, source caps (1, 6) and target caps (2, 3) give

$$P(u) = \frac{[6]_u}{[2]_u [3]_u} = u^2 - u + 1.$$

The denominator divides the numerator, so the zero-multiplicity obstruction disappears. The unique signed factor has a negative mass at h in the scaled version, and no universal kernel exists for any real common tilt. Transform-zero inclusion is therefore weaker than probability convolution divisibility.

5 Scope and attribution

The relevant repository question concerns arbitrary observation masks in Fabius conditioning [8]. Theorem 1.1 supplies a convolution criterion for all masks in the stated independent bounded-coordinate model. Theorem 2.1 completely resolves that universal order for reciprocal-integer geometric caps. Theorem 3.1 gives an exact finite algebraic criterion under its equal-cardinality and commensurability assumptions.

These are comparisons for one kernel required to work under the entire weight or threshold family. They do not decide every prescribed two-hypothesis comparison or the sharp asymptotics of overlap for a particular conditioning rule. The earlier companions [6, 7] treated the sufficient residue construction and the sharp individual-coordinate classification. The present proofs include the steps needed for overlapping and infinite masks.

Blackwell comparison and convolution criteria have substantial antecedents [1, 2]; exponential digit independence is classical [3], as are the Gaussian polynomial identities [4]. The note gives self-contained proofs in the stated sum-reweighted setting, with no general priority claim. These are ordinary mathematical proofs, not formalized or externally refereed results.

References

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